Reflection

In this assignment, I worked on two problems: "Path to Leaves" (Question 2) and "Missing Number in Range" (Question 3).

For Question 2, I approached the problem by implementing a breadth-first search (BFS) to explore all the paths from the root to the leaves of the binary tree. I used a queue to store the nodes and the current path from the root to that node.

As I encountered a leaf node, I added the path to the result. This solution ensures that we visit all paths in the tree while efficiently managing the traversal using BFS.

For Question 3, I suggested an optimization to my partner's solution. Initially, my partner used a boolean array to track seen numbers. While this approach works, I proposed a set-based solution, where we create a complete set of numbers from 0 to the maximum value in the list and subtract the numbers present in the input list. This is more efficient for larger datasets, as it avoids using an auxiliary array and eliminates the need to track all numbers individually.

During our collaboration, we reviewed each other's work through GitHub, providing feedback and suggestions to improve the clarity and efficiency of our solutions. It was rewarding to see how small changes can significantly optimize the code.

This experience helped me appreciate the value of peer review and collaborative problem-solving,

especially in coding assignments.